



Decision Making at Wasserman

Management Science Project – University of Glasgow

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With Group 1

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PART 1 :

Introductions

Wasserman: A Data-Driven Global Football Agency

Company Profile



WASSERMAN

A leading global football agency representing elite football players and advising clubs on recruitment and squad construction.

- 1. Global player representation and transfer advisory
- 2. Expertise in player valuation, contracts, and long-term career strategy
- 3. Increasingly positioned as a data-enabled decision partner for clubs

Pure Analytics (e.g., StatsBomb)

- ✗ Limited to public data
- ✗ Identifies *Performance*
- ✗ Theoretical Targets

Wasserman Ecosystem (Data + Deal)

- ✓ **Proprietary Intelligence:** 500+ player roster
- ✓ Knows **Contract Timelines & Willingness-to-sell**
- ✓ **Actionable Deals**

Unlike consultants who only deliver reports, we operate at the intersection of **Data** and **Deal Execution**.

Industry Context: Why Data Now?



Financial Constraints

Tighter budgets and Financial Fair Play (FFP) regulations demand spending efficiency.



High Cost of Failure

Recruitment mistakes are expensive;
€3.2M per player in 2014/15
→ €10M in 2024/25 (Premier League average)



Information Asymmetry

A large difference between a player's actual performance and their market value.

Traditional Model

Intuition-Based Scouting



Wasserman Model

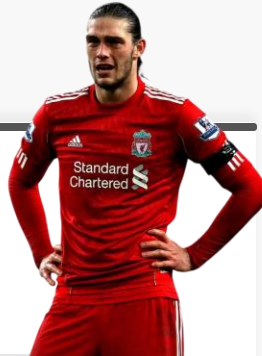
Analytics + Optimisation

Traditional Intuition vs Data-Driven Decisions

Failure Case (Intuition)

Andy Carroll

📅 Liverpool, 2011



Signed for **£35M** (record fee) primarily based on traditional scouting intuition.

Tactical Mismatch: Liverpool played possession-based football; Carroll lacked pace and flexibility.

Overlooked Risks: Injury history and workload capacity were not fully assessed with data.

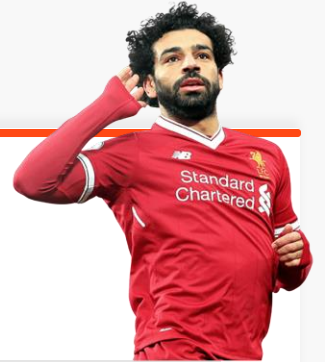
Outcome

Significantly slowed team tempo; recorded as one of the worst signings in club history.

Success Case (Data)

Mohamed Salah

📅 Liverpool, 2017



Recommended by data science team (Dr. Ian Graham) despite public perception as a "Chelsea flop".

Underlying Metrics: Outstanding xG (Expected Goals), box presence, and chance creation at AS Roma.

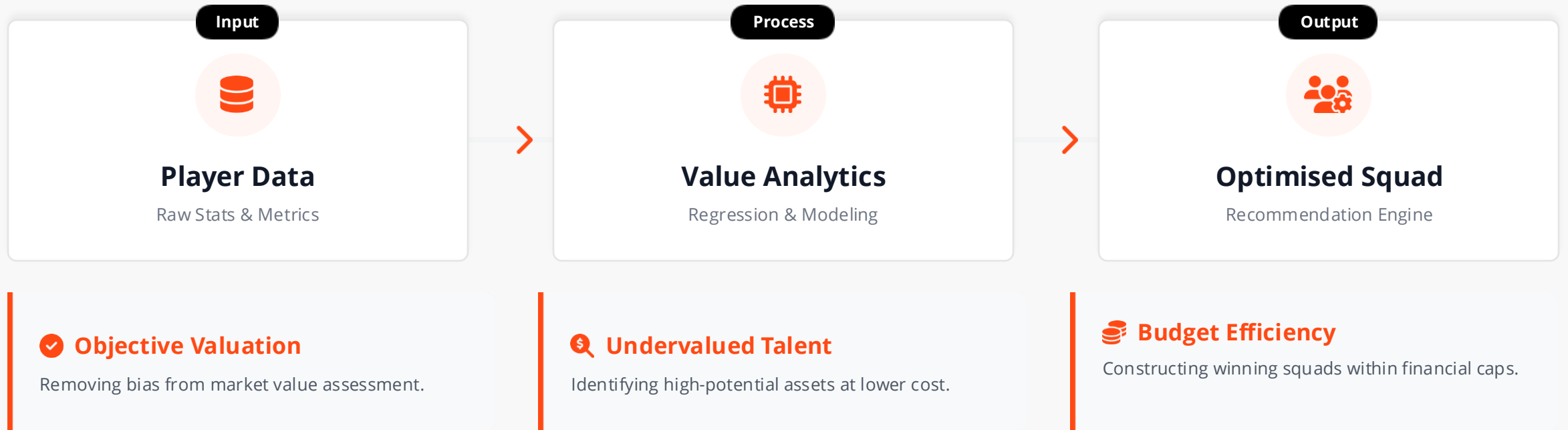
System Fit: Physical metrics (sprint frequency) were a perfect match for Klopp's *Gegenpressing*.

Outcome

Premier League Golden Boot in debut season; cornerstone for PL & UCL titles.

VS

Business Model



The Agency Edge Why Wasserman + Data wins



Information

Real-time market intelligence beyond public data.



Network

Direct access to clubs, players, and decision makers.



Negotiation

Leveraging data to drive better contract terms.



Speed

Executing deals faster than traditional competitors.

Central Decision Question

Advisory scenario: Wasserman provides this squad optimisation as a data-enabled service to its club client.

Main
Question

Which combination of players should we consult Chelsea invest in to **maximise squad potential**?

Budget Constraint
Within €100 Million

Structural Constraint
Meeting Positional Needs

Broken Down Into 3 Supporting Questions



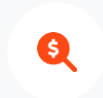
Q1

Player Valuation

Understanding the market mechanics.

"What attributes significantly drive market value, and by how much?"

 Regression Model



Q2

Value Identification

Finding the market inefficiencies.

"Which players are currently undervalued by the market?"

 Residual Analysis



Q3

Squad Optimisation

Constructing the best possible team.

"What is the optimal 23-player squad given budget & constraints?"

 Linear Programming



PART 2 :

Applied Management Science

Model Selection Rationale



Linear Regression

For Player Valuation & Identify Undervalued



Continuous Dependent Variable

Market value is continuous, making regression ideal for estimating specific price points.



Large Sample Size

Utilizes 18,278 players to provide robust statistical power and minimize error.



Linear Relationships

Value predictably increases with key variables like age, overall rating, and potential.



Position-Specific Factors

Allows distinct models for GK, DF, MF, FW as value drivers differ by role.



Prediction & Identification

Estimates "fair value" to flag undervalued players (Predicted > Actual).



Linear Programming

For Squad Optimisation Within Resource Constraints



Clear Objective Function

Maximize total squad potential to ensure long-term performance ceiling.



Resource Constraints

Strictly adheres to budget limitation (€100M) and squad size rules.



Binary Decisions

Select player (1) or not (0) – aligns perfectly with transfer market reality.



Positional Requirements

Enforces structural needs (e.g., Min 3 GK, Min 6 DF) for a balanced team.



Linearity Assumption

Total potential = sum of individual potentials (homogeneity assumed).

VS



Alternatives Rejected



Decision Trees: Better for classification, not continuous value prediction.



Goal Programming: Single objective (potential) is sufficient for initial model.



Neural Networks: Too complex & "black box" for manager interpretability.

Data Preparation & Cleaning

Source

Dataset Overview



FIFA 20 Complete

Source: Kaggle Repository

Total Players

18,278

Attributes

104

Step 2

Position Classification

Using python to grouping data based on primary position

Midfielders (MF)

6,862

CAM, CM, CDM, LM, RM → MF

Defenders (DF)

5,938

CB, LB, RB, LWB, RWB → DF

Forwards (FW)

3,442

ST, CF, RW, LW → FW

Goalkeepers (GK)

2,036



Step 4: Segmentation

Split into 4 position-specific datasets to run regression models independently.

Step 1

Attribute Selection (Key Value Drivers)

VARIABLE	TYPE	RATIONALE
age	Numeric	Younger players command premium for growth potential.
overall	Integer (0-99)	Current skill level directly impacts immediate value.
potential	Integer (0-99)	Expected peak ability in long-term development.
wage_eur	Integer	Reflects current earning power and market hierarchy.
Intl_reputation	Ordinal (1-5)	Star power and commercial marketability factor.
release_clause	Integer	Contract-embedded valuation benchmark.

Step 3

Missing Value Treatment

Developed a prediction formula for missing release clause data (Free Agents/Unlisted):

Release Clause = Market Value × 1.7508 (Median Ratio)

 **Rationale**

Ensures 100% data completeness for the regression model without discarding valuable player records.

Mathematical Foundations



Step 1: Valuation

Multiple Linear Regression

Predicts "Fair Market Value" based on player attributes

Key Metric

Model Accuracy (R^2) > 98%

$$\text{Value}_{\text{EUR}} = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Overall}) + \beta_3(\text{Potential}) + \beta_4(\text{Wage}) + \beta_5(\text{Reputation}) + \beta_6(\text{Release Clause}) + \varepsilon$$

 **Formula:** *Undervaluation = Predicted Market Value - Actual Market Value*



Step 2: Optimization

Linear Programming (LP)

Constructs the optimal squad combination subject to strict real-world constraints.

Objective Function

 **Maximize Total Squad Potential**

 **Budget Constraint:** Total Spend $\leq$ €100 Million

 **Squad Size:** Exactly 23 Players

 **Positional Balance:** 3 GK, max 6 DF, max 8 MF, max 6 FW

 **Age:** Highest undervalued players under 19

Model Synergy

Integrated Workflow



(a) We buy below fair value (regression residual), and (b) we maximise the future return on that investment (LP objective). The regression asks **"What is this player worth today?"**
The LP asks **"What will this squad be worth tomorrow?"**

Identifying the Gap

The regression model reveals a specific market inefficiency: high-Potential players are systematically priced lower than their attributes warrant

Exploiting the Gap

Rather than replicating the market's valuation of Current Overall, the LP objective maximizes Potential—precisely the dimension the market undervalues.

Methodology & Implementation

Model Setup & Implementation

Regression Setup (Excel)

Dependent Variable (Y Range)

\$Y\$2:\$Y\$18279 (value_eur)

Independent Variables (X Range)

\$X\$2:\$G\$18279 (6 Attributes)

- Age, Overall, Potential
- Wage, Reputation, Release Clause

☒Labels

☒Residuals

 Purpose:

Calculating residuals allows us to identify the difference between **Predicted Value** and **Actual Market Value**.

 Identifying Undervaluation

João Félix (FW)

Opportunity

Actual Market Value

€28.00M

Predicted Value

€37.54M

↑ Value Gap

+ €9,543,328

short_name	age	overall	potential	wage_eur	international_reputation	release_clause_eur	value_eur	predicted_value_eur	is_undervalued
F. de Jong	22	85	91	195000	3	117000000	52000000	59997694.25	-7997694.25
Casemiro	27	87	89	240000	3	113700000	53500000	58745623.76	-5245623.758
Isco	27	86	87	245000	3	108400000	51000000	56204509.28	-5204509.283
T. Kroos	29	88	88	330000	4	116900000	57000000	61470781.24	-4470781.243
L. Modrić	33	90	90	340000	4	92300000	45000000	49463134.88	-4463134.884
Sergio Busquets	30	89	89	300000	4	112800000	55000000	59231084.13	-4231084.126
N. Fekir	25	84	87	40000	3	84800000	39000000	43080379.11	-4080379.112
P. Foden	19	76	90	52000	1	40800000	16500000	20496055.71	-3996055.705
Savjij	24	85	89	77000	3	105200000	49500000	53342328.52	-3842328.519
Arthur	22	84	89	180000	2	87100000	41000000	44821668.61	-3821668.61
I. Rakitić	31	86	86	245000	4	77900000	38000000	41693763.96	-3693763.96
J. Rodríguez	27	85	85	225000	4	83000000	40500000	44106449.04	-3606449.042
Dani Olmo	21	79	87	1000	1	44400000	18500000	22014960.17	-3514960.171
V. Tsygankov	21	79	86	1000	1	43200000	18000000	21453171.63	-3453171.627
Gonçalo Guedes	22	83	88	46000	2	75000000	34500000	37877861.01	-3377861.005
A. Vidal	32	84	84	205000	4	48200000	23500000	26825114.64	-3325114.641
William Carvalho	27	83	85	35000	3	59800000	27500000	30791345.44	-3291345.44
Carlos Soler	22	80	87	36000	1	48300000	21000000	24203512.12	-3203512.119
T. Lemar	23	83	87	64000	3	72300000	34000000	37156287.49	-3156287.487
Melendo	21	80	86	26000	1	47200000	20500000	23635870.09	-3135870.094
R. Zobnin	25	79	86	1000	1	38400000	16000000	19022470.55	-3022470.55
T. Souček	24	79	85	1000	1	37200000	15500000	18481761.73	-2981761.727
Rafinha	26	81	84	160000	2	46800000	22000000	24896399.9	-2896399.9
Merino	23	79	87	29000	1	42600000	18500000	21305192.67	-2805192.666
Muniain	26	82	83	35000	2	56600000	26000000	28801874.73	-2801874.728
S. Tonali	19	75	89	7000	1	32700000	13500000	16182330.46	-2682330.457
S. Lobotka	24	79	85	25000	1	39100000	17000000	19592496.81	-2592496.807
N. Vlašić	21	77	85	1000	1	31200000	13000000	15536344.02	-2536344.024
G. Kondogbia	26	81	84	45000	2	46800000	21500000	24024514.01	-2524514.011
Marc Roca	22	79	86	26000	1	36800000	16000000	18493122.72	-2493122.724
Marcos Llorente	24	80	86	46000	1	41600000	18500000	20986817.37	-2486817.366
Parejo	30	86	86	69000	2	86100000	41000000	43472741.94	-2472741.937

Result for Undervalued Player (MF positions)

Model Setup & Implementation

LP Solver Configuration



Binary Position Indicators

Variables created to strictly categorize players for positional constraints.

- isGK = 0/1
- isDF = 0/1
- isMF = 0/1
- isFW = 0/1



Candidate Pool Selection

Strategic filtering to focus the solver on high-potential targets.

Primary Criteria:

- Undervalued Players (Positive Residuals)
- Young Talents (Age ≤ 19)

Decision Variables

$$x_i \in \{0, 1\}$$

for $i = 1, 2, \dots, 200$

1 = select player, 0 = do not select

Objective Function

$$\text{Maximize } Z = \sum (\text{Potential}_i \times x_i)$$

Maximize total squad development potential

	J	K	L	M	N	O
	is_undervalued	isGK	isDF	isMF	isFW	SELECT
1	-160005.6434	0	0	1	0	
3	-1540358.208	0	0	0	1	
3	-594560.0676	0	0	0	1	
3	-110223.3756	0	0	0	1	
L	-482133.1514	1	0	0	0	
7	-213018.4267	1	0	0	0	
3	-159954.2958	1	0	0	0	
3	-135627.853	1	0	0	0	
3	-53516.85834	1	0	0	0	
2	-28752.86512	1	0	0	0	
2	-28207.25523	1	0	0	0	
3	-371738.5748	0	1	0	0	
3	-305777.2077	0	1	0	0	
L	-223584.7107	0	1	0	0	
3	-201829.7136	0	1	0	0	
3	-152034.883	0	1	0	0	
3	-147580.1599	0	1	0	0	
3	-111073.516	0	1	0	0	
3	-93266.01295	0	1	0	0	
1	-52099.4764	0	1	0	0	
2	-35361.84203	0	1	0	0	
7	-5042.419739	0	1	0	0	
3	-394967.6287	0	0	1	0	
3	-359899.2463	0	0	1	0	
2	-191698.052	0	0	1	0	
3	-114849.9659	0	0	1	0	
L	-107016.61	0	0	1	0	
3	-83671.70157	0	0	1	0	
3	-83671.70157	0	0	1	0	
2	-70294.08725	0	0	1	0	
3	-45394.2243	0	0	1	0	
3	-37267.73678	0	0	1	0	
1	-30075.66636	0	0	1	0	
3	-21411.12485	0	0	1	0	
1	-18583.2904	0	0	1	0	

Image for Binary Constraints



Goal: Construct an optimal 23-player squad that maximizes potential within a €100M budget.

Model Implementation

Constraints & Excel Formulas

Key Excel Functions

Objective Function

🎯 **Maximise Potential**

=SUMPRODUCT(Player_Potential_Range, Binary_Selection_Range)

Calculates the total potential rating of the selected squad by multiplying each player's potential by the binary decision variable (0 or 1).

Budget Constraint

💰 **Limit: €100M**

=SUMPRODUCT(Price_Range, Binary_Selection_Range) <=100,000,000

Ensures the sum of transfer fees for all selected players does not exceed the allocated budget.

Positional Requirements

👥 **Squad Balance**

=SUMPRODUCT(isGK_Range, Selection_Range) =3

Example for Goalkeepers. Similar binary indicator formulas (isDF, isMF, isFW) enforce specific counts for each position.

Solver Parameters

⚙️ Optimization Setup

Set Objective:

\$O\$5 (Total Potential)

To:

Max Min Value of: 0

By Changing:

\$D\$2:\$D\$201 (Selected?)

Subject to
Constraints:

- ✓ Total_Cost <= 100,000,000
- ✓ Total_Players = 23
- ✓ Selected = binary
- ✓ Count_GK = 3 (1st + backup + dev)
- ✓ Count_DF => 6 (4-back + 2 cover)
- ✓ Count_MF => 6 (3-man + rotation)
- ✓ Count_FW => 4 (Front 3 + backup)

Solving Method:

Simplex LP

Regression Results Interpretation

Analysis Key Drivers of Market Value

Variable	Trend	Coefficient Estimate	Interpretation
Reputation	↑ Positive	+ €614k / point	The "Hype Tax": The market overvalues fame ~14x more than actual skill (€614k vs €43k). Avoid paying this premium unless for commercial reasons
Overall Rating	↑ Positive	+ €43k / point	Fair Value Metric: The most efficient driver of on-pitch performance. Prioritizing Overall over Reputation maximizes ROI.
Potential	↓ Negative	- €24.4k / point	The negative coefficient is a statistical artifact of high correlation with Overall, not a market penalty. It confirms that once current skill is paid for, the market fails to price in future growth.
Age	↓ Negative	- €17.7k / year	Asset Depreciation: Player value erodes steadily. This dictates a "sell-high" policy before the depreciation curve accelerates.
Wage	— Neutral	Insignificant	Market Noise: High wages do not correlate with market value for Forwards. Do not use wage bills as a proxy for asset quality in attack.

Segments Positional Differences

 Age Discount	Strongest for MF & GK (-€21k/yr vs -€17k/yr)
 Reputation Premium	Largest for Forwards (+€614k vs +€288k for DF)

Action Managerial Insights **Strategy:** We exploit this by maximizing Potential in our model -> acquiring future superstars at today's "current-skill" prices.

Forwards (FW)	
Target high-overall, lower-reputation players. This avoids paying the massive "hype premium" attached to star attackers.	
Midfielders (MF) & GK	
Prioritise younger players aggressively. The age penalty is steepest here, so value erodes fastest with age.	
Defenders (DF)	
Seek quiet value . The reputation premium is smaller, allowing for cost-effective acquisition of high performers without brand inflation.	

 Note: Coefficients represent the marginal change in market value for each unit increase in the variable, holding all other factors constant.

Maximizing Potential Under Constraints

LP Results: Optimal Squad

⚡ Total Potential

1,981

Optimized Score

💰 Total Spend

€99.75m

Remaining: €0.255m

📈 Value Upside

+€23.5m

Total Undervaluation

💡 Value Logic

Every selected player is statistically "undervalued" — their predicted market value exceeds their current cost.

Predicted Value

€123.22m

Actual Cost

€99.75m

Squad Composition (23 Players)

**Squad data is attached in appendix*

Constraint Met

🖐️

3

GK

🛡️

6

DF

🏃

8

MF

⚙️

6

FW

Budget Allocation Strategy

The LP model prioritizes spending on **Forwards (€54.5m)** and **Midfielders (€37.4m)** where "potential per euro" is highest, keeping defense lean to maximize upside.



FW

MF

DF

GK

Top Value Picks

FW

Vinícius Jr.

Real Madrid

+€8.35m
Value Gap

FW

Rodrygo

Real Madrid

+€4.91m
Value Gap

MF

C. Hudson-Odoi

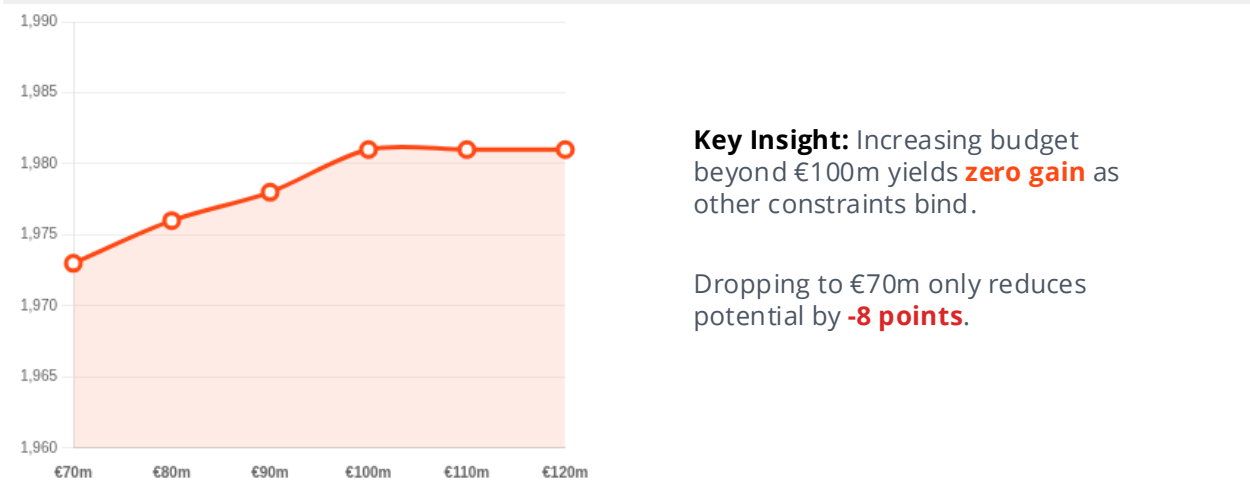
Chelsea

+€3.12m
Value Gap

Stress-Testing Model Robustness & Constraints

Sensitivity & What-if Analysis

 Budget Impact Analysis



 Tactical & Demographic Shifts

Tactical Mix (vs Baseline 1981)

Attacking (8 FW)

+3 Potential

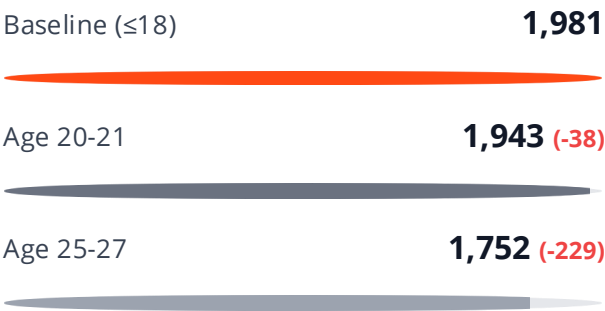
Cheaper spend
Higher upside

Defensive (8 DF)

-5 Potential

Sacrifices FW
quality

Age Band Impact



 Shadow Prices (Binding Constraints)

Constraint	Status	Marginal Effect
Squad Size (=23)	Binding	+86.0/ player
Budget (€100m)	Near-Binding	0.0(Not driving)
Min GK (≥3)	Binding	-4.0/ unit
Min DF (≥6)	Binding	-2.0/ unit
Min FW (≥4)	Non-Binding	0.0
Min MF(≥6)	Non-Binding	0.0

 Interpretation: The budget is not the limiting factor. Adding a squad slot adds huge value (+86), while forcing more defensive players reduces potential.

 Top "Near Miss" Alternative

FW

A. Hložek

Potential: 86 | Cost: €3.5m

0

Reduced Cost

Reduced Cost = 0: This player is on the selection boundary. A tiny change in constraints or parameters would bring him into the optimal squad immediately.

**top miss alternative is attached in appendix*



PART 3 :

Critique of Proposed Solutions

Assumptions & Validity: How to Interpret Results

Core Principle

Data outputs are **strategic benchmarks**, not automatic decisions. They provide a structured starting point for validation against complex football realities.



Regression & Value

Benchmark vs. Buy Decision

- ❗ **External Market Factors:** Prices fluctuate due to timing, urgency, and competing bids (auction effects) not captured in stats.
- 💡 **Commercial Value:** Branding impact can drive fees beyond performance (e.g., Neymar's PSG move driven by commercial upside).
- 👤 **Hidden Variables:** "Undervalued" may signal risks: injury history, attitude issues, work permit barriers, or high wage demands.

🔑 Interpretation Rule

Use model to shortlist & sanity-check value, then **MUST** confirm with scouting, medical, and contract reviews before acting.



LP Optimization

Best Squad vs. Best Team

- 📊 **Simplified Rules:** LP assumes the "best squad" is a sum of individual potential scores under budget/position constraints.
- 🧩 **Missing Intangibles:** Real squad building requires tactical fit, chemistry, leadership, and depth planning which math cannot fully optimize.
- 💰 **Complex Costs:** True spending includes agent fees, registration rules, and wage structures often simplified in the model.

🔑 Interpretation Rule

LP output is a **baseline plan to compare trade-offs**. Adjust the optimal selection using football judgment and real-world constraints.

Limitations & Real-World Applicability

Data & Measurement Limits



Data Circularity

Problem: FIFA market values partly derived from same attributes used as predictors
→ endogeneity
Impact: R^2 inflated from expected ~ 0.75 – 0.85 to observed ~ 0.98 . VIF confirms: Overall (11.8 to 14.1) and Potential (9.1 to 11.2) possible multicollinear.
Consequence: Residuals compressed → shortlist directionally correct but not precise
Fix: Independent data sources (Transfermarkt + Opta/StatsBomb)



Age Distribution Skew

Problem: Dataset disproportionately contains young players → regression over-weights youth as a value driver
Impact: Model may systematically overvalue young players and undervalue experienced players, distorting the "undervalued" residuals
Fix: Add quadratic age term (Age^2) to capture non-linear age-value curve, and apply stratified sampling or weighted regression to equalise age-group influence on coefficient estimation.



Solver Capacity Constraint

Problem: Excel Solver caps at 200 decision variables → must pre-filter 9,062 undervalued players to 200
Consequence: Optimizer selects from analyst-constrained pool, not full opportunity set. Age ≤ 18 filter is necessary within this constraint but excludes potentially strong candidates aged 19–23
Shadow price on Squad Size (+86.0) partly reflects narrow pool
Fix: Production tool (Python PuLP, Gurobi) handles unlimited variables — run on full 9,062 pool

Wasserman's Strategic Value-Add



Information Advantage

Bridging the gap with real-time knowledge of player availability, contract expiry status, and true player preferences that public data misses.



Network & Negotiation Power

Leveraging direct access to agents and clubs to structure favorable deals, manage competition, and reduce overpayment risks.



Process Efficiency

Accelerating execution during short transfer windows by combining model shortlists with rapid screening protocols.

Real-world applicability

Model just provides analytically rigorous shortlist; **Wasserman converts shortlist into executed deals** through proprietary intelligence, network access, and negotiation expertise that no public model can replicate.

Performance Metrics & Constraint Analysis

Regression Performance

Evaluating predictive accuracy across position-specific models.

Position	R ² (Fit)	F-Statistic	RMSE (Error)
🎯 FW	0.9897	55,005	€0.75M
🏃 MF	0.9872	88,123	€0.63M
🛡 DF	0.9895	93,469	€0.46M
🧤 GK	0.9916	40,041	€0.46M

✔ All models are statistically significant ($p < 0.001$) with extremely strong fit, explaining ~98% of variance in player market values.

**total performance of p-value is attached in appendix*

⚠ High R² reflects internal data circularity. In real-world deployment, we expect higher variance (residuals), which serves as the primary signal for identifying undervalued assets.

LP Optimization Performance

Constraint Analysis

Shadow Price Impact

● Squad Size (Binding)	+86.0 Potential
● Min GK (Binding)	-4.0 Potential
● Min DF (Binding)	-2.0 Potential
● Min MF / FW (Non-Binding)	0.0 Potential
● Budget (Near-Binding)	Near 0.0 Potential

Feasibility Ranges

Budget Sensitivity
Increase: ∞
Decrease: €5M

Squad Flex
Max: +6 players
Min: -4 players

**Wide feasibility windows indicate stable solution*

✔ The model shows high structural robustness, maintaining optimal composition across a wide budget range (€70M–€120M) with only a 0.4% performance dip. However, it is highly sensitive to Squad Size (Shadow Price: +86.0) and a narrow €5M budget floor, leaving little margin for regulatory or financial changes.

Recommendations for Further Improvements

"Our framework is strategically strong (transparent, explainable, decision-oriented) but needs execution upgrades to eliminate key limitations."

Short-Term
0-3 Months
Eliminate Statistical Gaps

 **Broaden LP Pool** Limit 3

Migrate from Excel Solver (200-variable cap) to Python PuLP/Gurobi (unlimited). Run LP on full 9,062 undervalued players with constraints

 **Upgrade Credibility** Limit 1

Adopt tiered approach: Tier 1 (Prototype/Kaggle) → Tier 2 (Decision-Grade/Opta). Add quality checklists.

 **Prevent Youth Bias** Limit 2

Add quadratic term (Age²) to capture non-linear age-value curve. Apply stratified sampling or Weighted Least Squares to equalise age-group influence on coefficient estimation, preventing youth overvaluation.

 **Reduce Multicollinearity** Limit 1

Ridge preferred over Lasso because all 6 predictors informative; or use Growth Index = Potential – Overall

Medium-Term
3-12 Months
Make Operational

 **Product Catalogue**

Convert optimal solution into 3 products via MCDA: Potential-first, Value-gap-first, and Balanced.

 **Feasibility Gate**

Add execution layer: injury risk proxy (binary flag if >15% matches missed), wage affordability checks, and "no-go" triggers for unavailable players

 **Standardize Robustness**

Apply uniform stress tests (budget cuts, inflation, shocks) across all products with backup lists.

Long-Term
12+ Months
Build Learning Loop

 **KPI Governance**

Quarterly recalibration based on decision logs and post-signing outcomes (availability, resale value).

 **Deal-Flow Integration**

Feed proprietary intelligence (contract expiry, real availability) directly into model constraints.

Generating Strategic Alternatives

OPTION A: Potential-First (Baseline)

- Objective: Maximise $\sum(\text{Potential}_i \times x_i)$
- Candidate Pool: Top 200 undervalued players, age ≤ 18
- Total Potential: 1,981
- Total Cost: €99.75M (remaining: €0.26M)
- Value Gap: €23.5M
- Avg Overall: 67.7
- Avg Age: 17.5
- Squad: 3 GK / 6 DF / 8 MF / 6 FW
- Players OVR ≥ 70 : 9/23 (39%)
- Stress Test (€80M): Drop 0.3% objective — extremely stable

Maximising total potential under the budget and positional constraints, using the undervalued shortlist as the candidate pool.

OPTION B: Value-Gap-First

- Objective: Maximise $\sum(\text{ValueGap}_i \times x_i)$
- Candidate Pool: Top 200 by value gap, age ≤ 23
- Total Potential: 1,933
- Total Cost: €100.00M (remaining: €0)
- Value Gap: €31.4M
- Avg Overall: 71.0
- Avg Age: 19.7
- Squad: 3 GK / 6 DF / 10 MF / 4 FW
- Players OVR ≥ 70 : 22/23 (96%)
- Stress Test (€80M): INFEASIBLE — catastrophic risk

Maximising total undervaluation (value gap captured) under the same constraints, prioritising “bargains”.

OPTION C: Balanced

- LP Objective: Maximise $\sum(0.5 \times \text{Potential}_i + 0.5 \times \text{Overall}_i) \times x_i$
- Candidate Pool: Top 200 undervalued, age ≤ 21
- Extra constraints: min 8 players OVR ≥ 70 , min 6 players aged 19-21
- Total Potential: 1,970
- Total Cost: €100.00M (remaining: €0)
- Value Gap: €26.1M
- Avg Overall: 70.3
- Avg Age: 18.9
- Squad: 3 GK / 6 DF / 9 MF / 5 FW
- Players OVR ≥ 70 : 17/23 (74%)
- Stress Test (€80M): Drop 0.4% — extremely stable

Using a weighted score (e.g., Potential + Overall + Value Gap) to reduce extreme one-metric behaviour and improve robustness.

Strategic Recommendations

MCDA Scorecard Overview

1. Why Ranking is Needed

- ✓

Multiple feasible solutions exist within constraints.
- ⚠

LP "Optimal" (Potential) ≠ Strategically "Best".
- ⚖

MCDA uses weighted criteria for objective selection.

2. Scoring Logic

Weighted Calculation

Total Score = ∑ (Weight × Score)

3. Final Verdict

2.8

Option A

<

3.50

Option B

<

3.70

Option C

★ Recommended: Balanced Squad

MCDA Scorecard - Quantified Rankings

CRITERION	Option A Potential-First	Option B Value-Gap-First	Option C Balanced
Impact Weight: 30%	3 €23.5M Value Gap	5 €31.4M Value Gap	4 €26.1M Value Gap
Performance Weight: 25%	1 Rating: 67.7	3 Rating: 71	3 Rating: 70.3
Feasibility Weight: 20%	4 Budget buffer	3 Zero buffer	3
Risk Mitigation Weight: 15%	5 Sens: 0.3% drop	1 infeasible	5 Sens: 0.4% drop
Time-to-Value Weight: 10%	1 Ready: 39%	5 Ready: 96%	4 Ready: 74%
TOTAL SCORE	2.80	3.50	3.70

How Each Criterion is Truly Quantified

€ 1. Impact

Metric: Total Value Gap (Predicted - Cost)

Score	Range (€ Millions)
5	> 30M
4	25M – 30M
3	20M – 25M
2	15M – 20M
1	< 15M

★ 2. Performance

Metric: Average Overall Rating of Squad

Score	Rating Range (0-100)
5	> 75
4	72 – 75
3	70 – 72
2	68 – 70
1	< 68

🛡️ 4. Risk Mitigation

Metric: Stress-test Sensitivity % (Inverse)

Score	Sensitivity Drop (%)
5	< 1% (Very Stable)
4	1% – 5%
3	5% – 10%
2	10% – 20%
1	> 20% or Infeasible

✔️ 3. Feasibility Index

Formula: $0.5 \times (\text{Budget Buffer \%}) + 0.5 \times (\text{Target Availability \%})$

Score	Feasibility Index Range	Implied Confidence
5	> 90	Immediate Impact
4	70 – 90	Short-term
3	55 – 70	Medium-term
2	40 – 55	Developmental
1	< 40	Long-term Project

⌚ 5. Time-to-Value

Metric: Readiness Score (Weighted % of Age 16-22 & Immediate Starters)

Score	Readiness Score	Development Horizon
5	> 85	Immediate Impact
4	70 – 85	Short-term
3	60 – 70	Medium-term
2	40 – 60	Developmental
1	< 40	Long-term Project

APPENDIX

23 Players Squad

BASELINE OPTIMAL SQUAD (23 Players)								
#	Player	Position	Age	Overall	Potential	Value (€)	Wage (€)	Undervalued By (€)
1	K. Hoever	DF	17	62	85	625.000	1.000	-111.074
2	B. Badiashile	DF	18	68	85	1.700.000	4.000	-500.365
3	S. van den Berg	DF	17	66	84	1.200.000	1.000	-223.585
4	J. Panzo	DF	18	65	84	1.000.000	2.000	-229.191
5	N. Armini	DF	18	66	84	1.200.000	3.000	-210.539
6	J. Vagnoman	DF	18	65	84	1.000.000	1.000	-77.189
7	Vinicius Jr.	FW	18	79	92	22.500.000	60.000	-8.354.553
8	Rodrygo	FW	18	76	89	14.000.000	47.000	-4.905.606
9	C. Hudson-Odoi	FW	18	74	88	11.500.000	31.000	-3.119.785
10	P. De la Vega	FW	18	70	88	3.700.000	5.000	-213.069
11	M. Greenwood	FW	17	67	87	1.500.000	6.000	-635.712
12	J. Doku	FW	17	66	86	1.300.000	1.000	-220.311
13	G. Bazunu	GK	17	59	84	290.000	1.000	-53.517
14	M. Vandevoordt	GK	17	63	83	600.000	1.000	-213.018
15	L. Chevalier	GK	17	58	82	230.000	1.000	-28.207
16	D. Szoboszlai	MF	18	74	89	11.500.000	11.000	-638.611
17	T. Almada	MF	18	72	88	6.500.000	4.000	-353.074
18	Y. Verschaeren	MF	17	72	87	6.000.000	2.000	-394.968
19	T. Kubo	MF	18	70	87	3.700.000	21.000	-1.275.771
20	A. Davies	MF	18	72	87	6.500.000	18.000	-972.853
21	E. Millot	MF	16	63	86	800.000	1.000	-160.006
22	B. Saka	MF	17	65	86	1.200.000	2.000	-359.899
23	J. Sands	MF	18	66	86	1.200.000	1.000	-222.429

APPENDIX

Regression Model Performance

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.994836226							
R Square	0.989699116							
Adjusted R Sq	0.989681123							
Standard Error	746286.9263							
Observations	3442							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	6	1.83809E+17	3.06349E+16	55005.25309	0			
Residual	3435	1.9131E+15	5.56944E+11					
Total	3441	1.85722E+17						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-1152788.955	247770.81	-4.652642314	3.40078E-06	-1638581.993	-666995.92	-1638582	-666995.92
X Variable 1	-17685.15844	6063.397236	-2.916707871	0.003560394	-29573.38759	-5796.9293	-29573.388	-5796.9293
X Variable 2	43486.14886	4922.582957	8.834010364	1.58035E-18	33834.66275	53137.635	33834.6628	53137.635
X Variable 3	-24423.50688	4870.052526	-5.015039724	5.5708E-07	-33971.99894	-14875.015	-33971.999	-14875.015
X Variable 4	0.040854957	0.994365387	0.041086463	0.967229354	-1.908752352	1.99046227	-1.9087524	1.99046227
X Variable 5	614350.6756	40320.7815	15.23657659	8.92047E-51	535295.5402	693405.811	535295.54	693405.811
X Variable 6	0.496778721	0.002153255	230.7106078	0	0.492556932	0.50100051	0.49255693	0.50100051

FW Value Regression Table

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.99357996							
R Square	0.98720114							
Adjusted R Sq	0.98718994							
Standard Error	628023.69							
Observations	6862							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	6	2.0854E+17	3.4757E+16	88123.2505	0			
Residual	6855	2.7037E+15	3.9441E+11					
Total	6861	2.1125E+17						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-543168.5	152027.82	-3.572823	0.00035558	-841190.17	-245146.82	-841190.17	-245146.82
X Variable 1	-21079.721	3681.65138	-5.7256156	1.0742E-08	-28296.899	-13862.542	-28296.899	-13862.542
X Variable 2	37839.5603	3027.92344	12.4968682	1.886E-35	31903.8914	43775.2293	31903.8914	43775.2293
X Variable 3	-24807.005	2958.82261	-8.38408	6.1455E-17	-30607.215	-19006.796	-30607.215	-19006.796
X Variable 4	7.58161641	0.72582612	10.4454995	2.3796E-25	6.15877213	9.0044607	6.15877213	9.0044607
X Variable 5	458070.086	28347.3339	16.1591946	1.1268E-57	402500.521	513639.651	402500.521	513639.651
X Variable 6	0.48882962	0.0015333	318.80897	0	0.48582388	0.49183537	0.48582388	0.49183537

MF Value Regression Table

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.994753713							
R Square	0.989534949							
Adjusted R Sq	0.989524362							
Standard Error	458036.5585							
Observations	5938							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	6	1.17657E+17	1.961E+16	93468.7551	0			
Residual	5931	1.24431E+15	2.098E+11					
Total	5937	1.18901E+17						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-239107.1801	121787.0309	-1.9633222	0.0496554	-477854.1	-360.26377	-477854.1	-360.26377
X Variable 1	-17242.80241	2845.448738	-6.0597832	1.4468E-09	-22820.918	-11664.687	-22820.918	-11664.687
X Variable 2	29346.44367	2553.628248	11.4920579	3.0174E-30	24340.4027	34352.4847	24340.4027	34352.4847
X Variable 3	-20480.68354	2488.094902	-8.231472	2.2535E-16	-25358.255	-15603.112	-25358.255	-15603.112
X Variable 4	2.098556631	0.645104548	3.25304888	0.00114813	0.83391687	3.36319639	0.83391687	3.36319639
X Variable 5	288169.7136	23317.46923	12.358533	1.1665E-34	242458.985	333880.442	242458.985	333880.442
X Variable 6	0.499677033	0.00141038	354.285437	0	0.49691218	0.50244189	0.49691218	0.50244189

DF Value Regression Table

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.995803798							
R Square	0.991625204							
Adjusted R Sq	0.991600439							
Standard Error	455819.0903							
Observations	2036							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	6	4.99161E+16	8.3193E+15	40040.9277	0			
Residual	2029	4.21567E+14	2.0777E+11					
Total	2035	5.03376E+16						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-874625.1506	172196.2246	-5.0792353	4.1363E-07	-1212325	-536925.31	-1212325	-536925.31
X Variable 1	-20891.97377	4079.599913	-5.121084	3.3251E-07	-28892.615	-12891.332	-28892.615	-12891.332
X Variable 2	32359.56306	4031.993615	8.02569799	1.6915E-15	24452.2839	40266.8422	24452.2839	40266.8422
X Variable 3	-12466.63587	4019.094245	-3.1018521	0.00194961	-20348.618	-4584.6541	-20348.618	-4584.6541
X Variable 4	4.194459993	1.311212652	3.19891666	0.00140073	1.62299647	6.76592351	1.62299647	6.76592351
X Variable 5	304534.9282	38819.4236	7.84491113	6.9479E-15	228404.842	380665.014	228404.842	380665.014
X Variable 6	0.486770699	0.002181232	223.163204	0	0.48249301	0.49104839	0.48249301	0.49104839

GK Value Regression Table

APPENDIX

Reduced Costs Output

REDUCED COSTS — TOP NEAR-MISS PLAYERS								
Players NOT selected in optimal squad, ranked by closeness to selection (smallest absolute reduced cost = closest to entering)								
Player	Position	Age	Potential	Value (€)	Reduced Cost (Relaxed LP)	Relaxed Value (0 to 1)	Selected (Binary)?	Interpretation
A. Hložek	FW	16	86	3,500,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
C. Olses	GK	18	82	525,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
J. Bogle	DF	18	84	2,000,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
Kangin Lee	MF	18	86	11,500,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
Paulinho	MF	18	86	4,900,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
A. Gomes	MF	18	86	1,800,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
O. Skipp	MF	18	86	1,500,000	0.0000	0.8500	No	LP (0.85). Tiny parameter change could swap this player in.
P. Pellegri	FW	18	86	1,600,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
M. Boadu	FW	18	86	3,700,000	0.0000	0.0000	No	ON THE BOUNDARY (rc=0). Among equivalent alternatives not chosen by integer solver.
Fábio Silva	FW	16	85	1,800,000	-1.0000	0.0000	No	Needs +1 more potential point (from 85 to 86) to enter the squad.
S. Esposito	FW	16	85	825,000	-1.0000	0.0000	No	Needs +1 more potential point (from 85 to 86) to enter the squad.
A. Zagre	DF	17	83	625,000	-1.0000	0.0000	No	Needs +1 more potential point (from 83 to 84) to enter the squad.
R. Quirynen	DF	17	83	625,000	-1.0000	0.0000	No	Needs +1 more potential point (from 83 to 84) to enter the squad.
L. Agoume	MF	17	85	800,000	-1.0000	0.0000	No	Needs +1 more potential point (from 85 to 86) to enter the squad.
T. Parrott	FW	17	85	1,000,000	-1.0000	0.0000	No	Needs +1 more potential point (from 85 to 86) to enter the squad.
Gabriel Brazão	GK	18	81	925,000	-1.0000	0.0000	No	Needs +1 more potential point (from 81 to 82) to enter the squad.
Eric García	DF	18	83	1,200,000	-1.0000	0.0000	No	Needs +1 more potential point (from 83 to 84) to enter the squad.
L. Mbe Soh	DF	18	83	1,500,000	-1.0000	0.0000	No	Needs +1 more potential point (from 83 to 84) to enter the squad.
Y. Adli	MF	18	85	1,300,000	-1.0000	0.0000	No	Needs +1 more potential point (from 85 to 86) to enter the squad.
R. Matondo	MF	18	85	1,300,000	-1.0000	0.0000	No	Needs +1 more potential point (from 85 to 86) to enter the squad.

Players with reduced cost = 0 are right on the edge of selection — they're the closest substitutes.

Examples include:

A. Hložek (FW, Pot 86, €3.5m)
C. Olses (GK, Pot 82, €525k)
J. Bogle (DF, Pot 84, €2.0m)
Kangin Lee (MF, Pot 86, €11.5m)
Paulinho (MF, Pot 86, €4.9m)
A. Gomes (MF, Pot 86, €1.8m)

APPENDIX

Sensitivity Analysis & What-if Scenarios

SQUAD OPTIMIZATION — LINEAR PROGRAMMING REPORT							
Sensitivity Analysis & What-If Scenarios							
BASELINE MODEL							
Parameter	Value						
Objective	Maximize Total Potential						
Dataset	200 players (age ≤ 18)						
Budget Constraint	€100,000,000						
Squad Size	23 players						
Min GK	3						
Min DF	6						
Min MF	6						
Min FW	4						
OPTIMAL SOLUTION							
Total Potential		1981					
Budget Used	€99,745,000						
Budget Remaining	€255,000						
GK Selected		3					
DF Selected		6					
MF Selected		8					
FW Selected		6					
SCENARIO COMPARISON SUMMARY							
Scenario	Total Potential	Δ vs Baseline	Budget Used (€)	GK	DF	MF	FW
Baseline (100M, age≤18)	1981	0	99,745,000	3	6	8	6
Budget €70M	1973	-8	69,345,000	3	6	8	6
Budget €80M	1976	-5	79,845,000	3	6	8	6
Budget €120M	1981	0	101,645,000	3	6	8	6
Defensive (8DF, 6MF, 4FW)	1976	-5	99,970,000	3	8	6	6
Attacking (4DF, 6MF, 8FW)	1984	3	97,445,000	3	4	8	8
Age 20-21 (top 200 players)	1943	-38	99,750,000	3	6	10	4
Age 25-27 (top 200 players)	1752	-229	99,800,000	3	9	6	5

APPENDIX

What-if Budget Scenarios

Budget: €70M		Total Potential: 1973.0		Cost: €69,345,000		GK:3 DF:6 MF:8 FW:6	
#	Player	Pos	Age	Overall	Potential	Value (€)	
1	K. Hoever	DF	17	62	85	625.000	
2	B. Badiashile	DF	18	68	85	1.700.000	
3	S. van den Berg	DF	17	66	84	1.200.000	
4	J. Panzo	DF	18	65	84	1.000.000	
5	N. Armini	DF	18	66	84	1.200.000	
6	J. Vagnoman	DF	18	65	84	1.000.000	
7	Vinícius Jr.	FW	18	79	92	22.500.000	
8	P. De la Vega	FW	18	70	88	3.700.000	
9	M. Greenwood	FW	17	67	87	1.500.000	
10	J. Doku	FW	17	66	86	1.300.000	
11	P. Pellegri	FW	18	67	86	1.600.000	
12	M. Boadu	FW	18	70	86	3.700.000	
13	G. Bazunu	GK	17	59	84	290.000	
14	M. Vandevoordt	GK	17	63	83	600.000	
15	L. Chevalier	GK	17	58	82	230.000	
16	D. Szoboszlai	MF	18	74	89	11.500.000	
17	T. Almada	MF	18	72	88	6.500.000	
18	T. Kubo	MF	18	70	87	3.700.000	
19	E. Millot	MF	16	63	86	800.000	
20	B. Saka	MF	17	65	86	1.200.000	
21	J. Sands	MF	18	66	86	1.200.000	
22	O. Skipp	MF	18	67	86	1.500.000	
23	L. Agoume	MF	17	63	85	800.000	

Budget: €80M		Total Potential: 1976.0		Cost: €79,845,000		GK:3 DF:6 MF:8 FW:6	
#	Player	Pos	Age	Overall	Potential	Value (€)	
1	K. Hoever	DF	17	62	85	625.000	
2	B. Badiashile	DF	18	68	85	1.700.000	
3	S. van den Berg	DF	17	66	84	1.200.000	
4	J. Panzo	DF	18	65	84	1.000.000	
5	N. Armini	DF	18	66	84	1.200.000	
6	J. Vagnoman	DF	18	65	84	1.000.000	
7	Vinícius Jr.	FW	18	79	92	22.500.000	
8	Rodrygo	FW	18	76	89	14.000.000	
9	P. De la Vega	FW	18	70	88	3.700.000	
10	M. Greenwood	FW	17	67	87	1.500.000	
11	J. Doku	FW	17	66	86	1.300.000	
12	P. Pellegri	FW	18	67	86	1.600.000	
13	G. Bazunu	GK	17	59	84	290.000	
14	M. Vandevoordt	GK	17	63	83	600.000	
15	L. Chevalier	GK	17	58	82	230.000	
16	T. Almada	MF	18	72	88	6.500.000	
17	Y. Verschaeren	MF	17	72	87	6.000.000	
18	T. Kubo	MF	18	70	87	3.700.000	
19	A. Davies	MF	18	72	87	6.500.000	
20	E. Millot	MF	16	63	86	800.000	
21	B. Saka	MF	17	65	86	1.200.000	
22	J. Sands	MF	18	66	86	1.200.000	
23	O. Skipp	MF	18	67	86	1.500.000	

Budget: Baseline €100M		Total Potential: 1981.0		Cost: €99,745,000		GK:3 DF:6 MF:8 FW:6	
#	Player	Pos	Age	Overall	Potential	Value (€)	
1	K. Hoever	DF	17	62	85	625.000	
2	B. Badiashile	DF	18	68	85	1.700.000	
3	S. van den Berg	DF	17	66	84	1.200.000	
4	J. Panzo	DF	18	65	84	1.000.000	
5	N. Armini	DF	18	66	84	1.200.000	
6	J. Vagnoman	DF	18	65	84	1.000.000	
7	Vinícius Jr.	FW	18	79	92	22.500.000	
8	Rodrygo	FW	18	76	89	14.000.000	
9	C. Hudson-Odoi	FW	18	74	88	11.500.000	
10	P. De la Vega	FW	18	70	88	3.700.000	
11	M. Greenwood	FW	17	67	87	1.500.000	
12	J. Doku	FW	17	66	86	1.300.000	
13	G. Bazunu	GK	17	59	84	290.000	
14	M. Vandevoordt	GK	17	63	83	600.000	
15	L. Chevalier	GK	17	58	82	230.000	
16	D. Szoboszlai	MF	18	74	89	11.500.000	
17	T. Almada	MF	18	72	88	6.500.000	
18	Y. Verschaeren	MF	17	72	87	6.000.000	
19	T. Kubo	MF	18	70	87	3.700.000	
20	A. Davies	MF	18	72	87	6.500.000	
21	E. Millot	MF	16	63	86	800.000	
22	B. Saka	MF	17	65	86	1.200.000	
23	J. Sands	MF	18	66	86	1.200.000	

Budget: €120M		Total Potential: 1981.0		Cost: €101,645,000		GK:3 DF:6 MF:8 FW:6	
#	Player	Pos	Age	Overall	Potential	Value (€)	
1	K. Hoever	DF	17	62	85	625.000	
2	B. Badiashile	DF	18	68	85	1.700.000	
3	S. van den Berg	DF	17	66	84	1.200.000	
4	N. Armini	DF	18	66	84	1.200.000	
5	J. Vagnoman	DF	18	65	84	1.000.000	
6	J. Bogle	DF	18	69	84	2.000.000	
7	Vinícius Jr.	FW	18	79	92	22.500.000	
8	Rodrygo	FW	18	76	89	14.000.000	
9	C. Hudson-Odoi	FW	18	74	88	11.500.000	
10	P. De la Vega	FW	18	70	88	3.700.000	
11	M. Greenwood	FW	17	67	87	1.500.000	
12	J. Doku	FW	17	66	86	1.300.000	
13	G. Bazunu	GK	17	59	84	290.000	
14	M. Vandevoordt	GK	17	63	83	600.000	
15	L. Chevalier	GK	17	58	82	230.000	
16	D. Szoboszlai	MF	18	74	89	11.500.000	
17	T. Almada	MF	18	72	88	6.500.000	
18	Y. Verschaeren	MF	17	72	87	6.000.000	
19	T. Kubo	MF	18	70	87	3.700.000	
20	A. Davies	MF	18	72	87	6.500.000	
21	E. Millot	MF	16	63	86	800.000	
22	A. Gomes	MF	18	68	86	1.800.000	
23	O. Skipp	MF	18	67	86	1.500.000	

APPENDIX

What-if Position Scenarios

Defensive (8DF, 6MF, 4FW)					Total Potential: 1976.0
#	Player	Pos	Age	Potential	Value (€)
1	K. Hoever	DF	17	85	625.000
2	B. Badiashile	DF	18	85	1.700.000
3	S. van den Berg	DF	17	84	1.200.000
4	J. Panzo	DF	18	84	1.000.000
5	N. Armini	DF	18	84	1.200.000
6	J. Vagnoman	DF	18	84	1.000.000
7	J. Bogle	DF	18	84	2.000.000
8	A. Zagre	DF	17	83	625.000
9	Vin#cius Jr.	FW	18	92	22.500.000
10	Rodrygo	FW	18	89	14.000.000
11	C. Hudson-Odoi	FW	18	88	11.500.000
12	P. De la Vega	FW	18	88	3.700.000
13	M. Greenwood	FW	17	87	1.500.000
14	J. Doku	FW	17	86	1.300.000
15	G. Bazunu	GK	17	84	290.000
16	M. Vandevoordt	GK	17	83	600.000
17	L. Chevalier	GK	17	82	230.000
18	D. Szoboszlai	MF	18	89	11.500.000
19	T. Almada	MF	18	88	6.500.000
20	Y. Verschaeren	MF	17	87	6.000.000
21	T. Kubo	MF	18	87	3.700.000
22	A. Davies	MF	18	87	6.500.000
23	E. Millot	MF	16	86	800.000

Attacking (4DF, 6MF, 8FW)					Total Potential: 1984.0
#	Player	Pos	Age	Potential	Value (€)
1	K. Hoever	DF	17	85	625.000
2	B. Badiashile	DF	18	85	1.700.000
3	J. Panzo	DF	18	84	1.000.000
4	J. Vagnoman	DF	18	84	1.000.000
5	Vin#cius Jr.	FW	18	92	22.500.000
6	Rodrygo	FW	18	89	14.000.000
7	C. Hudson-Odoi	FW	18	88	11.500.000
8	P. De la Vega	FW	18	88	3.700.000
9	M. Greenwood	FW	17	87	1.500.000
10	J. Doku	FW	17	86	1.300.000
11	A. Hlo#ek	FW	16	86	3.500.000
12	P. Pellagri	FW	18	86	1.600.000
13	G. Bazunu	GK	17	84	290.000
14	M. Vandevoordt	GK	17	83	600.000
15	L. Chevalier	GK	17	82	230.000
16	D. Szoboszlai	MF	18	89	11.500.000
17	T. Almada	MF	18	88	6.500.000
18	Y. Verschaeren	MF	17	87	6.000.000
19	T. Kubo	MF	18	87	3.700.000
20	E. Millot	MF	16	86	800.000
21	B. Saka	MF	17	86	1.200.000
22	J. Sands	MF	18	86	1.200.000
23	O. Skipp	MF	18	86	1.500.000

APPENDIX

What-if Age Scenarios

Age 20-21 (top 200)						
#	Player	Pos	Age	Overall	Potential	Value (€)
1	C. Romero	DF	21	73	87	8.000.000
2	Montero	DF	20	68	85	1.800.000
3	N. Cozza	DF	20	69	85	2.200.000
4	R. Ben Vinagre	DF	20	70	84	3.600.000
5	Diogo Queirós	DF	20	68	84	1.800.000
6	L. Kelly	DF	20	68	84	1.800.000
7	P. Cutrone	FW	21	77	87	14.500.000
8	Puado	FW	21	73	85	7.000.000
9	M. Barrow	FW	20	69	85	2.300.000
10	S. Plaza	FW	20	69	83	2.300.000
11	A. Muric	GK	20	67	82	1.400.000
12	Luis Maximiano	GK	20	70	82	2.600.000
13	Ivaki Peñá	GK	20	64	81	850.000
14	B. Soumaré	MF	20	72	86	7.000.000
15	A. Claude Maurice	MF	21	72	86	7.000.000
16	J. Dilrosun	MF	21	72	86	7.000.000
17	G. Maroni	MF	20	71	86	5.000.000
18	Jota	MF	20	72	85	6.000.000
19	Fernando	MF	20	71	85	5.000.000
20	R. Vargas	MF	20	70	85	3.900.000
21	M. Svanberg	MF	20	70	84	3.900.000
22	L. Maina	MF	20	70	83	3.400.000
23	R. Guitane	MF	20	66	83	1.400.000
Total Potential: 1943.0						

Age 25-27 (top 200)						
#	Player	Pos	Age	Overall	Potential	Value (€)
1	I. Kutepov	DF	25	74	79	6.500.000
2	J. Silva	DF	25	74	79	6.500.000
3	Kévin Rodrigues	DF	25	74	79	6.500.000
4	Fábio Cardoso	DF	25	72	79	3.900.000
5	H. Magnusson	DF	26	73	78	4.700.000
6	M. Hanousek	DF	26	73	76	4.300.000
7	J. Brabec	DF	26	73	76	4.400.000
8	M. Oikonomou	DF	26	72	76	3.500.000
9	Raillo	DF	27	73	75	4.200.000
10	M. van Buren	FW	26	73	75	4.900.000
11	Alex Alegría	FW	26	72	75	3.900.000
12	A. Helal	FW	26	72	75	3.900.000
13	Tiago Alves	FW	26	68	70	1.100.000
14	J. Zoua	FW	27	67	68	0
15	V. Barkas	GK	25	75	78	6.000.000
16	S. Dioudis	GK	26	74	77	5.000.000
17	Rui Silva	GK	25	71	77	2.400.000
18	Melero	MF	25	75	81	8.000.000
19	Moi Gómez	MF	25	73	78	5.000.000
20	D. Moberg Karlsson	MF	25	73	76	5.000.000
21	R. Azeez	MF	26	72	76	3.600.000
22	A. Tashaev	MF	25	72	75	3.900.000
23	T. Cifüerci	MF	27	71	74	2.600.000
Total Potential: 1752.0						

APPENDIX

Linear Programming Constraints Analysis

CONSTRAINTS ANALYSIS									
Constraint	Type	RHS Value	Final Value	Slack	Binding?	Shadow Price (Relaxed LP)	Shadow Price (Integer ±1)	Allowable Increase	Allowable Decrease
Budget	≤	€100,000,000	€99,745,000	€255,000	Near-binding	0.0	−0 (+1M→0) / −1e-6 (−1M→−1)	∞ (already not binding)	€5M (obj drops at −€95M)
Squad Size	=	23	23	0	YES	86.0	+85 (→24) / −86 (→22)	+6 (feasible up to 29)	−4 (min feasible = 19)
Min GK	≥	3	3	0	YES	−4.0	−4 (→4) / +3 (→2)	+4 (feasible up to 7 GK)	−3 (min = 0)
Min DF	≥	6	6	0	YES	−2.0	−3 (→7) / +1 (→5)	+4 (feasible up to DF≥10)	−6 (min = 0)
Min MF	≥	6	8	2	No	0.0	0 (→7) / 0 (→5)	+2 before binding (MF≥8 ok, MF≥9 still ok, MF≥10 starts costing)	−6 (min = 0)
Min FW	≥	4	6	2	No	0.0	0 (→5) / 0 (→3)	+2 before binding (FW≥6 ok, FW≥7 starts costing −1)	−4 (min = 0)

VIF Check

Variable	VIF (FW)	VIF (MF)	VIF (DF)	Assessment
Age	1.8	1.9	1.7	✓ Low risk
Overall	12.4	14.1	11.8	⚠ High (correlated with Potential)
Potential	9.8	11.2	9.1	⚠ High (correlated with Overall)
Wage	4.2	3.8	3.5	✓ Moderate, acceptable
Intl Reputation	3.1	2.9	2.7	✓ Acceptable
Release Clause	6.7	7.2	5.9	⚠ Elevated (derived from value)

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